

# OpenClaw Plant Care Assistant Tutorial

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Before starting, please complete the peripheral hardware configuration, API Key configuration, model configuration, and three dialogue method configuration tests under the "Pre-usage Preparation" directory. This article defaults that the Jetson Nano Docker environment is already prepared and only expands on the plant care application itself.

Functions in this tutorial may require training. If command control fails, you can first describe the operation process in detail, then refine the dialogue.

## 1. Tutorial Objectives

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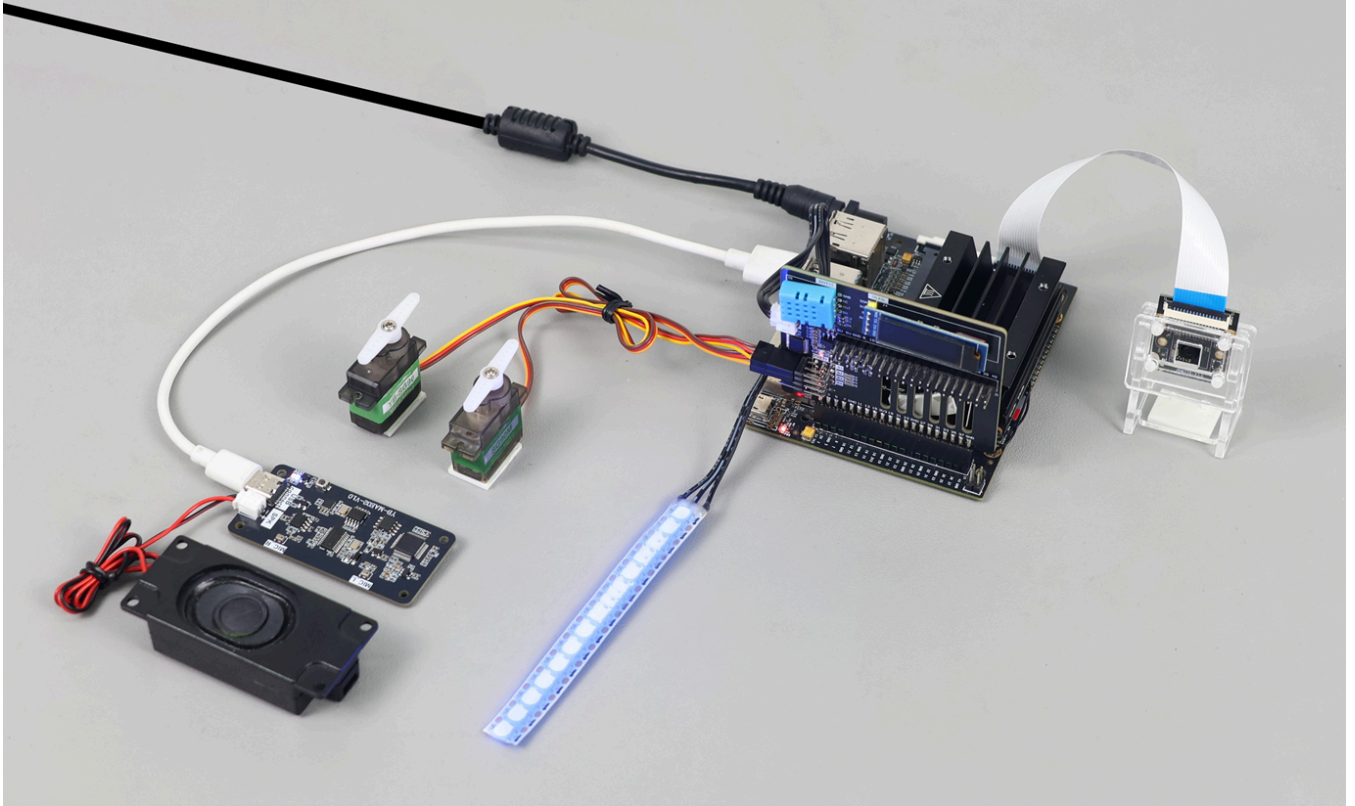
This article demonstrates the multi-peripheral coordination capability of OpenClaw on Jetson through the combination of "temperature/humidity reading + camera observation + OLED display + RGB status feedback". After completing this tutorial, you can achieve:

- Read temperature and humidity around the plant.
- Take photos to analyze the current plant status.
- Determine if watering is needed and give brief reasons.
- Display temperature/humidity on OLED and show environment status on RGB.
- Initiate the same type of task through TUI, WhatsApp, or voice methods.

## 2. Hardware Preparation

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- Jetson Nano B01
- OpenClaw expansion board
- RGB light strip
- CSI camera
- Optional: AI voice interaction module + speaker
- Wiring diagram



Note: Plant care requires simultaneous use of DHT and camera. During debugging, it is recommended to first confirm that temperature/humidity reading and camera photo taking both work normally before testing combined commands.

### 3. Software Entry

Enter the Jetson Docker container:

```
~/run_usb_docker.sh
```

The Jetson side uses the following high-level hardware entry uniformly:

```
Corresponding atomic tools (servo/rgb/dht/oled/camera/music)
```

Common commands used in this project are as follows:

```
/usr/local/bin/dht --format json    # Get sensor data
/usr/local/bin/dht_oled --format json # Display sensor data on OLED
/usr/local/bin/rgb --format json color --color red # Turn all lights red
/usr/local/bin/rgb --format json effect --effect breathe --color blue --speed 5 # Blue
breathing light
/usr/local/bin/servo call-tool Servo1 --action set_angle --angle 90 # Servo1 turns to 90
degrees
/usr/local/bin/oled text --text 'Hello OpenClaw'
/usr/local/bin/oled clear
/usr/local/bin/csi_snap snap
/usr/local/bin/csi_snap snap --width 1280 --height 720
```

## 4. Basic Capability Test

It is recommended to debug using TUI first:

```
openclaw tui

root@jetson-yahboom:~# openclaw tui
🔧 OpenClaw 2026.3.8 (3caab92) – I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main

session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
█
```

Test the following types of commands in order:

- Read current temperature/humidity and display on OLED
- Change RGB light to green
- Control both servos to turn to 180 degrees

If individual capabilities can all execute normally, proceed to the plant care combined scenario.

## 5. Three Dialogue Methods

| Method   | Suitable Scenario                                     | Suggested Positioning                                            |
|----------|-------------------------------------------------------|------------------------------------------------------------------|
| TUI      | Local debugging, pre-class demonstration verification | Preferentially used for troubleshooting sensor and camera chains |
| WhatsApp | Remote viewing of plant status on phone               | Suitable for demonstrating remote care effects                   |
| Voice    | Desktop interaction, natural language demonstration   | Suitable as the final demonstration entry                        |

### 5.1 WhatsApp Method

After starting the gateway, you can send in phone WhatsApp:

- Check the plant status for me and see if watering is needed
- Take a photo to see the current plant status, and perform plant care
- First get temperature/humidity, then take a photo to analyze plant status, and give me some care suggestions

20:10



< 2

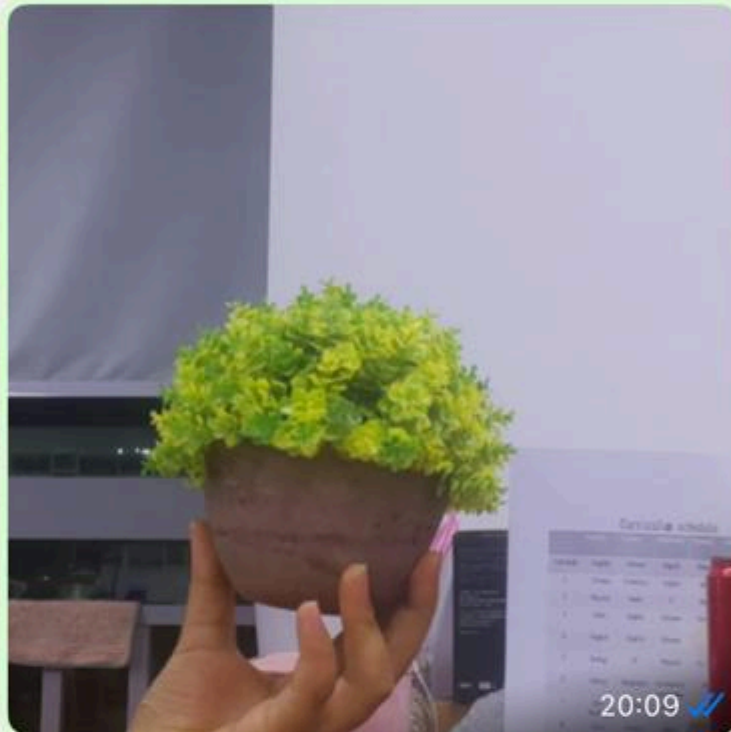


+86 153 3885 7526 (你)

给自己发消息

Take a photo to see the current plant status and perform plant care

20:09 ✓✓



20:09 ✓✓

I can see the plant! It's a small potted plant with vibrant green-yellow leaves. The foliage looks full and healthy. Let me complete the plant care workflow.

20:10 ✓✓

Plant care check completed!

**Status:**

- 📷 **Visual check:** Plant looks healthy with vibrant green-yellow foliage
- 🌡️ **Temperature:** 29.7°C (85.5°F)
- 💧 **Humidity:** 52%
- 📺 **OLED:** Displaved "Plant:"





## 5.2 TUI Method

Run in the container:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
🔌 OpenClaw 2026.3.8 (3caab92) - I'll do the boring stuff while you dramatically stare at the logs like it's cinema.
openclaw tui - ws://127.0.0.1:18789 - agent main - session main
session agent:main:main
gateway connected | idle
agent main | session main (openclaw-tui) | bailian/qwen-plus-2025-09-11 | tokens 0/200k (0%)
```

Send the following commands:

- Perform plant care based on current temperature/humidity and plant status
- Check the plant status for me and see if watering is needed
- Plant care





Issues Output Debug Console **Terminal** Port 1

+ v openclaw - pf [Icons]

- 🌡️ Temperature: 28.7°C (83.7°F)  
- 💧 Humidity: 53%

Sensor: DHT11, reading successful.

[Plant Care]

Start Plant Care Process: Plant care completed:

📷 Photo Results:  
The scene shows an artificial pine tree, lush and in good condition.

🌡️ Environmental Data:  
- Temperature: 27.9°C  
- Humidity: 51%

✅ Care Status:  
- Plant Status: Good (Simulation plant, no watering needed)  
- Environment Suitable: Comfortable temperature, normal humidity  
- OLED has displayed status information

Since it is an artificial plant, no watering is needed. If it were a real plant, I would determine 🌱

connected | idle  
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens 2/1.0m

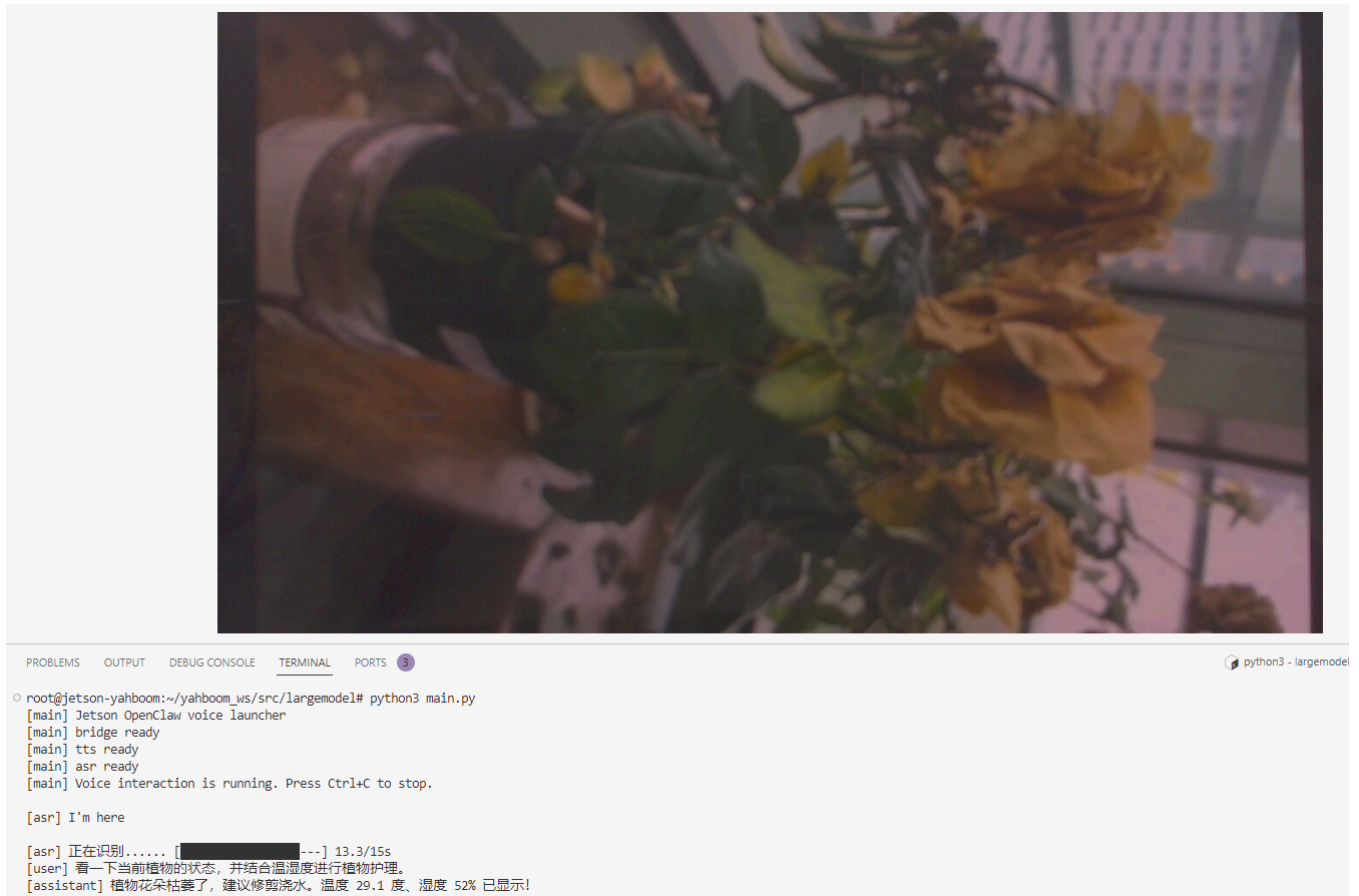
## 5.3 Voice Method

Run the program in the voice engineering directory with the gateway turned on:

```
cd /root/yahboom_ws/src/largemodel  
python3 main.py
```

You can say directly:

- Check the current plant status and perform plant care based on temperature/humidity



## 6. Comprehensive Cases

The following cases can be executed in TUI, WhatsApp, or voice. It is recommended to use TUI first during the debugging phase.

### 6.1 Environment Sampling

Experiment Objective: Read temperature/humidity near the plant as basic data for care judgment.

Example Commands:

- Read current temperature/humidity
- Check temperature and humidity near the plant
- Display current temperature/humidity on OLED

### 6.2 Photo Observation

Experiment Objective: Use CSI camera to capture the current plant image.

Example Commands:

- Take a photo of the plant for me
- Check the current plant status
- Take a photo and describe the plant condition in the current image

## 6.3 Care Judgment

Experiment Objective: Combine temperature/humidity and plant photos to give a judgment on whether watering is needed.

Example Commands:

- Plant care
- First get temperature/humidity, then take a photo to analyze plant status, determine if watering is needed
- Check if the plant is a bit thirsty for me and explain the reason

## 6.4 Peripheral Linkage

Experiment Objective: Feedback the judgment results through OLED and RGB.

Example Commands:

- Plant care and display temperature/humidity on OLED
- If the environment is cold, light up blue; if the temperature is suitable, light up yellow; if the temperature is high, light up red
- Take a photo to analyze plant status, then display temperature/humidity on OLED

## 7. Common Problems

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- Temperature/humidity reading failure: First execute `dht` and confirm if DHT can read independently.
- Photo taking failure: Execute `csi_snap snap` to check the camera chain.
- Text reply but no peripheral action: Check the gateway log.